Answer key

Practice Test 5 (State/National level) · 50 questions · 36 with checked app keys

Each answer lists short steps, the vault note (Question Types folder) and the app keys. Every key line was run in the app and its screen value matched the computed answer (check_keys.py). Accept numeric answers within about 2% with the right unit.

Multiple-choice answers

1	2	3	4	5	6	7	8	9	10	11	12	13	14
D	D	D	C	C	B	D	A	C	C	C	B	A	A

15
B

Section I: Multiple choice

1. D. Spectroscopy: naming elements from the light they give off

1. Kirchhoff and Bunsen built the spectroscope (about 1859-1860).
2. With it they found the elements cesium and rubidium.

Note: A1 History/Kirchhoff
App: no app: fact or concept

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2. D. Methane

1. Volta collected flammable "marsh air" from Lake Maggiore: methane.
2. He also invented the electrophorus (1775).

Note: A1 History/Volta
App: no app: fact or concept

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3. D. Military engineer

1. Coulomb graduated from the military engineering school at Mezieres (1761).
2. He built forts for the French army before his electrical work (1785).

Note: A1 History/Coulomb
App: no app: fact or concept

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4. C. Cork

1. Queen's College, Cork (now University College Cork), Ireland, 1849.
2. Lincoln is where he was born (trap). His Laws of Thought came out in 1854.

Note: A1 History/Boole
App: no app: fact or concept

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5. C. A is negative and B is positive

1. The positive rod pulls electrons toward A, the near sphere.
2. Separating them traps the charges: A has extra electrons (negative), B has lost them (positive).
3. This is charging by induction: no charge moved from the rod.

Note: A2 Static Electricity and Coulomb/Charging Methods and Electroscopes
App: no app: fact or concept

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6. B. At the sharpest (most pointed) end

1. On a conductor, charge piles up where the surface curves most sharply.
2. That is why lightning rods and corona discharge use sharp points.
3. Inside the metal the field is zero, so no charge sits in the center.

Note: A2 Static Electricity and Coulomb/Electric Field and Potential
App: no app: fact or concept

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7. D. For the same power the current is smaller, so the I^2R loss in the lines is much smaller

1. P = VI: higher V means lower I for the same power.
2. Line loss = I^2R: half the current gives a quarter of the loss.

Note: A3 DC and AC/AC Characteristics
App: no app: fact or concept

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8. A. The two hot wires (each is 120 V from neutral, on opposite sides)

1. A US home gets split-phase power: two hot legs, each 120 V to neutral.
2. Between the two hot legs the voltage is 240 V. A double-pole breaker feeds both.

Note: A3 DC and AC/Household Wiring
App: no app: fact or concept

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9. C. 4

1. Same mass and metal: same volume, so A × L is the same.
2. Twice the length: half the area.
3. R = ρL/A: 2 × 2 = 4 times.

Note: A4 Current Voltage Resistance Power Energy/Resistance and Resistivity
App: no app: fact or concept

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10. C. A small, low-voltage circuit switches a separate high-power circuit, and the two are electrically isolated

1. The coil circuit and the contact circuit share no wire.
2. A few tens of mA in the coil can switch amps at 120 V safely.

Note: A6 Switches Relays Fuses Breakers GFCI/Relays
App: no app: fact or concept

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11. C. A ⊕ B ⊕ C

1. Each flip toggles the light, so the light shows whether an odd number of switches are 1.
2. Odd parity = A XOR B XOR C (3-input XOR).

Note: A6 Switches Relays Fuses Breakers GFCI/3-Way and 4-Way Switching
App: `ta^b^c` → `F = A'B'C + A'BC' + AB'C' + ABC` *(checked)*
App: `ta^b^c` → `This is a 3-input XOR gate` *(checked)*

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12. B. It compares the hot and neutral currents, which does not need a ground wire

1. A GFCI trips when hot and neutral currents differ by about 5 mA.
2. That test uses only hot and neutral. The outlet must be labeled "No equipment ground".

Note: A6 Switches Relays Fuses Breakers GFCI/GFCI
App: no app: fact or concept

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13. A. Good

1. A good silicon diode shows about 0.5-0.7 V forward and OL in reverse.
2. Shorted: about 0 V both ways. Open: OL both ways.

Note: A7 Diodes and LEDs/Testing Diodes
App: no app: fact or concept

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14. A. Gets wider, and only a tiny leakage current flows

1. Reverse bias pulls carriers away from the junction: the depletion region widens.
2. Only a very small leakage current flows until breakdown.

Note: A7 Diodes and LEDs/PN Junction and I-V Curve
App: no app: fact or concept

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15. B. 4

1. G1 = NAND(A, B).
2. G2 = NAND(A, G1), G3 = NAND(B, G1).
3. F = NAND(G2, G3) = A XOR B: 4 gates.

Note: A8 Digital Logic/Universal Gates (NAND and NOR)
App: `t-(-(a-(ab))-(b-(ab)))` → `F = A'B + AB'` *(checked)*

Section II: Short answer and calculations

16. 0.69 N, straight toward –y (toward the x-axis)

1. Distance from each charge: r = $\sqrt{(0.20^2 + 0.30^2)}$ = 0.3606 m.
2. Each pull: k(2.0e-6)(3.0e-6)/r^2 = 8.99e9 × 6.0e-12 / 0.13 = 0.4149 N (attraction).
3. By symmetry the x parts cancel.
4. Each y part = F × (0.30 / r) = 0.3452 N. Total = 2 × that = 0.69 N, toward –y.

Note: A2 Static Electricity and Coulomb/Coulombs Law
App: `Q 2u 0 0, 2u .4 0, -3u .2 .3` → `F on q3: 690.3 mN at -90 deg` *(checked)*
App: `Q 2u 0 0, 2u .4 0, -3u .2 .3` → `F on q3: 690.3 mN at -90 deg` *(checked)*

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17. (a) 2.7 × 10^2 V; (b) 2.7 × 10^{-7} J

1. Potential is a scalar: add kq/r for each charge (with signs).
2. V = 8.99e9 × (5.0e-9 – 2.0e-9) / 0.10 = 269.7 V = 2.7 × 10^2 V.
3. W = qV = 1.0e-9 × 269.7 = 2.7 × 10^{-7} J.

Note: A2 Static Electricity and Coulomb/Electric Field and Potential
App: `Q 5n 0, -2n .2 =.1` → `V at x=0.1: 269.6 V (F on q = qE)` *(checked)*
App: `f q3n r.1 t1n` → `j Potential energy U = 269.6 nJ` *(checked)*

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18. 1.0 N, repel (A: +4.5 µC, C: +2.25 µC)

1. A and B share +9.0 µC: each +4.5 µC.
2. B and C share +4.5 µC: each +2.25 µC.
3. F = 8.99e9 × 4.5e-6 × 2.25e-6 / 0.30^2 = 1.0114 N = 1.0 N, repel.

Note: A2 Static Electricity and Coulomb/Charge Basics
App: `q a4.5u b2.25u r.3` → `f Force (+ repel) = 1.011 N` *(checked)*

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19. 0.83 Ω

1. E = I(R + r) for each load.
2. E = 1.200(5.00 + r) = 6.000 + 1.200 r.
3. E = 0.750(8.50 + r) = 6.375 + 0.750 r.
4. Subtract: 0.450 r = 0.375 (only 2 significant figures are left after 6.375 − 6.000), so r = 0.83 Ω.

Note: A5 DC Circuits/Internal Resistance and Power Transfer
App: `j a1 b−1.2 c6 d1 e−0.75 f6.375` → `y y (I2 or V2) = 0.83333` *(checked)*

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20. 7.00 V

1. E = 6.000 + 1.200 × 0.8333 = 7.00 V.
2. Check: 0.750 × (8.50 + 0.8333) = 7.000 V.

Note: A5 DC Circuits/Internal Resistance and Power Transfer
App: `j a1 b−1.2 c6 d1 e−0.75 f6.375` → `x x (I1 or V1) = 7` *(checked)*
App: `i e7.0000 r0.8333 l5` → `i Current = 1.2 A` *(checked)*

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21. 6 in series × 4 in parallel = 24 cells; 222 Wh

1. Series: 22.0 / 3.70 = 5.95, round up to 6 (pack 22.2 V).
2. Parallel: 10.0 Ah / 2.50 Ah = 4 strings.
3. Cells = 6 × 4 = 24.
4. Energy = 22.2 V × 10.0 Ah = 222 Wh.

Note: A3 DC and AC/Batteries and Cells
App: `p s6 p4 v3.7 c2500` → `w Pack energy = 222 Wh` *(checked)*
App: `p s6 p4 v3.7 c2500` → `a Pack voltage = 22.2 V` *(checked)*

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22. 15.6 A

1. $I_{rms} = P/V_{rms}$ = 1320 / 120 = 11.0 A.
2. $I_{peak} = \sqrt{2} \times I_{rms}$ = 15.6 A.

Note: A3 DC and AC/AC Characteristics
App: `o p1320 v120` → `i Current = 11 A` *(checked)*
App: `w r11` → `p Peak (amplitude) = 15.56 V` *(checked)*

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23. 8.0 A; about 13 times (normal 0.625 A)

1. Cold: I = 120 / 15 = 8.00 A.
2. Hot (normal): I = P/V = 75/120 = 0.625 A.
3. Ratio = 8.00 / 0.625 = 12.8, about 13 (2 significant figures).

Note: A4 Current Voltage Resistance Power Energy/Current and Charge
App: `o v120 r15` → `i Current = 8 A` *(checked)*
App: `o p75 v120` → `i Current = 625 mA` *(checked)*

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24. 944 W

1. Energy = 112.20 / 0.165 = 680 kWh.
2. Time = 30 × 24 = 720 h.
3. P = 680 kWh / 720 h = 0.9444 kW = 944 W.

Note: A4 Current Voltage Resistance Power Energy/Power Energy and Cost
App: `e c112.2 r.165 h24 d30` → `p Power = 944.4 W` *(checked)*

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25. (a) red, yellow, white, brown, brown; (b) 4.99 kΩ

1. (a) Digits 2, 4, 9 = red, yellow, white. 249 × 10 = 2490: multiplier brown. 1% = brown.
2. (b) 4-digit code: three digits, then the number of zeros: 499 × 10 = 4990 ohm = 4.99 kohm.

Note: A5 DC Circuits/Resistor Color Codes
App: `c2.49k` → `5-band: red yellow white brown brown` *(checked)*
App: `c4991` → `If SMD code: 4991 = 4.99 kohm` *(checked)*

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26. 0.500 A; 16.7% low (true 0.600 A)

1. Without the meter: I = 6.00 / 10.0 = 0.600 A.
2. With the meter: I = 6.00 / (10.0 + 2.00) = 0.500 A.
3. Error = (0.600 − 0.500) / 0.600 = 16.7% low.

Note: A5 DC Circuits/Meters and Measurement
App: `N V1-0 6 1-2 2 2-0 10` → `V1-0 6: 500 mA out of +, gives 3 W` *(checked)*

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27. 5.40 Ω

1. Inject current at the center, take it out at corner 1.
2. By symmetry, corners 2 and 6 are at the same voltage, and corners 3 and 5 are too.
3. Join the equal nodes: A = 2+6, B = 3+5. Then C-A = R/2, C-B = R/2, A-B = R/2, 1-A = R/2, B-4 = R/2, C-4 = R, C-1 = R.
4. Node 4 is a side path: C-4-B = 1.5R, in parallel with C-B (R/2): 0.375R.
5. C to A: R/2 in parallel with (R/2 + 0.375R): 0.318R. C to 1: R in parallel with (R/2 + 0.318R): 0.45R.
6. R = 0.45 × 12.0 = 5.40 Ω.

Note: A5 DC Circuits/Hard Resistor Networks
App: `N1-2-3-4-5-6-1 12 1-7 2-7 3-7 4-7 5-7 6-7 =7-1` → `R(7-1) = 5.4 ohm` *(checked)*

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28. 396 Ω

1. The 470 ohm has 15.0 − 6.00 = 9.0 V: I = 9.0 / 470 = 19.149 mA.
2. The 1.50 k takes 6.00 / 1500 = 4.000 mA.
3. R takes 15.149 mA, so R = 6.00 / 0.015149 = 396 Ω.

Note: A5 DC Circuits/Series and Parallel Resistance
App: `N V1-0 15 1-2 470 2-0 1.5k 2-0 x V2=6` → `x = 396.1 ohm (R2-0)` *(checked)*

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29. 20 lamps

1. Microwave: 1150 / 120 = 9.583 A.
2. Left over: 20 − 9.583 = 10.417 A.
3. Each lamp: 60 / 120 = 0.500 A. 10.417 / 0.500 = 20.83.
4. Round DOWN: 20 lamps (lamp 21 would pass 20 A).

Note: A6 Switches Relays Fuses Breakers GFCI/Circuit Breakers
App: `F f20-1150/120 v120 p60` → `Max loads: 20 (load 21 blows the fuse)` *(checked)*

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30. 26.3 mA

1. The zener holds the load at 5.10 V.
2. Series resistor: (12.00 − 5.10)/220 = 31.36 mA.
3. Load: 5.10 / 1000 = 5.10 mA.
4. Zener: 31.36 − 5.10 = 26.3 mA.

Note: A7 Diodes and LEDs/Special Diodes
App: `N V1-0 12 1-2 220 D2-0 5.1 2-0 1k` → `D2-0 5.1: ON 26.26 mA 2>0 133.9 mW` *(checked)*

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31. F = B′C′ + B′D′ + A′C′D

1. Quad 0, 1, 8, 9 (B = 0, C = 0): B'C'.
2. Quad 0, 2, 8, 10 (the four corners, B = 0, D = 0): B'D'.
3. Minterm 5 pairs with 1 (A = 0, C = 0, D = 1): A'C'D.
4. F = B'C' + B'D' + A'C'D.

Note: A8 Digital Logic/Boolean Algebra and Simplification
App: `t abcd m0,1,2,5,8,9,10` → `F = A'C'D + B'C' + B'D'` *(checked)*

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32. 110010010 = 402

1. 11011011 = 219, 10110111 = 183.
2. 219 + 183 = 402.
3. 402 = 256 + 128 + 16 + 2 = 110010010 (9 bits: the carry makes a new bit).

Note: A8 Digital Logic/Binary Numbers and Addition
App: `x11011011+10110111` → `sum bin 110010010` *(checked)*

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33. S = 0010, C_{out} = 1

1. 11 + 6 + 1 = 18.
2. 18 = binary 10010: 5 bits.
3. The low 4 bits are S = 0010; the 5th bit is the carry-out, 1.

Note: A8 Digital Logic/Adders
App: `x1011+0110+1` → `sum bin 10010` *(checked)*

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34. I_0 = C, I_1 = C′, I_2 = 0, I_3 = 1

1. AB = 00: rows 0 (F = 0) and 1 (F = 1): F = C.
2. AB = 01: rows 2 (1) and 3 (0): F = C'.
3. AB = 10: rows 4 and 5 both 0: I_2 = 0.
4. AB = 11: rows 6 and 7 both 1: I_3 = 1.

Note: A8 Digital Logic/Multiplexers and Decoders
App: `t abc m1,2,6,7` → `F = A'B'C + AB + BC'` *(checked)*

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35. 51.8 mW

1. Current (same in the series path): I = 0.282 / 47.0 = 6.00 mA.
2. P = V × I = 8.64 × 0.00600 = 51.8 mW.

Note: A9 Hands-On and Meters/Task - Power Supplied
App: `o v8.64 i.282/47` → `p Power = 51.84 mW` *(checked)*

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36. 102 Ω

1. Red: I = (5.00 − 1.90)/150 = 20.67 mA.
2. Green: R = (5.00 − 2.90)/I = 2.10 / 0.02067 = 102 Ω.
3. The nearest standard value is 100 ohm.

Note: A9 Hands-On and Meters/Task - Equal LEDs
App: `l s5 f1.9 r150` → `i LED current = 20.67 mA` *(checked)*
App: `l s5 f2.9 i20.667m` → `r Resistor = 101.6 ohm` *(checked)*

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37. 150 + (220 || 68) = 201.9 Ω

1. 220 || 68 = 220 × 68 / 288 = 51.94 ohm.
2. Add 150 in series: 201.9 ohm.
3. Error = +0.97%: inside 1%.

Note: A9 Hands-On and Meters/Task - Target Resistance
App: `h 68 150 220 470=200` → `201.9 (+0.972%) 150+(220||68)` *(checked)*
App: `s150+220/68` → `Req = 201.9 ohm` *(checked)*

Section III: Circuit problems

Problem A: two batteries, one reversed (Kirchhoff's laws) (Questions 38–41)

E1 = 24.0 V has its + plate at the TOP. E2 = 9.00 V has its + plate at the BOTTOM. R1 = 8.00 Ω, R2 = 4.00 Ω, R3 = 16.0 Ω. Let I_1 be the current in R1 flowing toward node A and I_2 the current in R2 flowing toward node A.

38. 2.79 A

1. KCL: the current down through R3 is $I_1 + I_2$.
2. Left loop: 24.0 = 8.00 I_1 + 16.0($I_1 + I_2$), so 24 I_1 + 16 I_2 = 24.
3. Right loop, B up through E2: going from + to – the voltage DROPS 9.00 V. So 0 = –9.00 – 4.00 I_2 – 16.0($I_1 + I_2$): 16 I_1 + 20 I_2 = –9.
4. Determinant 24 × 20 – 16 × 16 = 224. I_1 = (480 + 144)/224 = 2.79 A.

Note: A5 DC Circuits/Kirchhoff Analysis (States-Nationals)
App: `j a24 b16 c24 d16 e20 f-9` → `x x (I1 or V1) = 2.7857` *(checked)*
App: `N V1-0 24 1-2 8 V0-3 9 3-2 4 2-0 16` → `V1-0 24: 2.786 A out of +, gives 66.86 W` *(checked)*

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39. I_2 = –2.68 A: 2.68 A flows from A through R2 toward E2

1. I_2 = (24 × (–9) – 16 × 24)/224 = –600/224 = –2.68 A.
2. The minus sign: the real current flows away from A, through R2, and down through E2 (in at its – plate).

Note: A5 DC Circuits/Kirchhoff Analysis (States-Nationals)
App: `j a24 b16 c24 d16 e20 f-9` → `y y (I2 or V2) = -2.6786` *(checked)*
App: `N V1-0 24 1-2 8 V0-3 9 3-2 4 2-0 16` → `V0-3 9: 2.679 A out of +, gives 24.11 W` *(checked)*

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40. 0.107 A (A to B); V_{AB} = 1.71 V

1. $I_3 = I_1 + I_2$ = 2.7857 – 2.6786 = 0.1071 A.
2. Better: use V_A = 12/7 V from the node equation below, so I_3 = (12/7)/16 = 3/28 = 0.107 A (subtracting the rounded 2.79 and 2.68 would lose a digit).
3. V_{AB} = 16.0 × 0.1071 = 1.71 V.
4. Node check: (V – 24)/8 + (V + 9)/4 + V/16 = 0 gives V = 12/7 V.

Note: A5 DC Circuits/Kirchhoff Analysis (States-Nationals)
App: `N V1-0 24 1-2 8 V0-3 9 3-2 4 2-0 16` → `R2-0 16: 1.714 V 107.1 mA 2>0 183.7 mW` *(checked)*
App: `N V1-0 24 1-2 8 V0-3 9 3-2 4 2-0 16` → `V1 = 24 V V2 = 1.714 V` *(checked)*

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41. 24.1 W

1. The current 2.6786 A enters E2 at its − plate and leaves at its + plate.
2. So E2 gives power: P = 9.00 × 2.6786 = 24.1 W.
3. Check: E1 gives 24.0 × 2.7857 = 66.86 W; total 90.96 W = I^2R sum 90.96 W.

Note: A5 DC Circuits/Kirchhoff Analysis (States-Nationals)
App: `N V1–0 24 1–2 8 V0–3 9 3–2 4 2–0 16` → `V0–3 9: 2.679 A out of +, gives 24.11 W` *(checked)*

Problem B: Wheatstone bridge with a galvanometer (Questions 42–45)

A 10.0 V ideal battery feeds the bridge below. P is between 220 Ω (top) and 330 Ω (bottom). Q is between 470 Ω (top) and R_x (bottom). The galvanometer G (50.0 Ω) joins P and Q.

42. 705 Ω

1. Balanced: the ratios match on both sides: 220/330 = 470/R_x.
2. R_x = 470 × 330 / 220 = 705 Ω.

Note: A5 DC Circuits/Hard Resistor Networks
App: `W a220 b330 c470` → `x Rx (bottom right) = 705 ohm` *(checked)*

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43. 0.563 V

1. V_P = 10.0 × 330/550 = 6.000 V.
2. V_Q = 10.0 × 560/1030 = 5.437 V.
3. $V_P - V_Q$ = 0.563 V.

Note: A5 DC Circuits/Dividers and Bulb Brightness
App: `N V1–0 10 1–2 220 2–0 330 1–3 470 3–0 560 V2–3` → `V2 – V3 = 563.1 mV` *(checked)*

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44. 1.29 mA from P to Q

1. Node P: (V_P − 10)/220 + V_P/330 + ($V_P - V_Q$)/50 = 0.
2. Node Q: (V_Q − 10)/470 + V_Q/560 + ($V_Q - V_P$)/50 = 0.
3. Solve the two equations: V_P = 5.8301 V, V_Q = 5.7658 V.
4. I = (5.8301 − 5.7658)/50 = 1.29 mA, from P to Q.

Note: A5 DC Circuits/Kirchhoff Analysis (States-Nationals)
App: `N V1–0 10 1–2 220 2–0 330 1–3 470 3–0 560 2–3 50` → `R2–3 50: 64.35 mV 1.287 mA 2>3 82.82 uW` *(checked)*

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45. 28.0 mA

1. Current in 220 ohm: (10 − 5.8301)/220 = 18.954 mA.
2. Current in 470 ohm: (10 − 5.7658)/470 = 9.009 mA.
3. Total = 28.0 mA.

Note: A5 DC Circuits/Kirchhoff Analysis (States-Nationals)
App: `N V1–0 10 1–2 220 2–0 330 1–3 470 3–0 560 2–3 50` → `V1–0 10: 27.96 mA out of +, gives 279.6 mW` *(checked)*

Problem C: NOR-only logic (Questions 46–48)

Three NOR gates: G1 = NOR(A, B), G2 = NOR(B, C), and F = NOR(G1, G2).

46. F = (A + B)(B + C) = B + AC

1. F = ((A + B)' + (B + C)')'.
2. De Morgan: F = (A + B)'' · (B + C)'' = (A + B)(B + C).
3. Expand: AB + AC + B + BC = B + AC (B absorbs AB and BC).

Note: A8 Digital Logic/De Morgan and Bubble Pushing
App: `t-(-(a+b)+-(b+c))` → `F = AC + B` *(checked)*

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47. F = Σm(2, 3, 5, 6, 7)

1. B = 1: rows 2, 3, 6, 7.
2. AC = 1: rows 5, 7.
3. Together: 2, 3, 5, 6, 7.

Note: A8 Digital Logic/Truth Tables and Boolean Expressions
App: `t-(-(a+b)+-(b+c))` → `m(2,3,5,6,7)` *(checked)*

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48. F = ΠM(0, 1, 4) = (A + B)(B + C)

1. The maxterms are the rows with F = 0: 0, 1, 4 (000, 001, 100).
2. This matches the NOR-NOR form: the output is the product of the two OR terms.

Note: A8 Digital Logic/Truth Tables and Boolean Expressions
App: `t-(-(a+b)+-(b+c))` → `M(0,1,4)` *(checked)*

Problem D: diode logic (Questions 49–50)

Inputs A and B are each 0 V or 5.0 V. D1 and D2 are silicon diodes (0.7 V when on), with their anodes at OUT. R = 4.7 kΩ goes from +5.0 V to OUT. Nothing else is connected to OUT.

49. AND gate

1. Any input at 0 V turns its diode on and pulls OUT down to 0.7 V (a logic 0).
2. Only when A and B are both 5.0 V are both diodes off, so R pulls OUT up to 5.0 V.
3. Output high only for A AND B.

Note: A7 Diodes and LEDs/Diode Applications
App: no app: fact or concept

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50. V_{OUT} = 0.7 V; I = 0.91 mA

1. D1 (cathode at 0 V) is on: V_{OUT} = 0 + 0.7 = 0.7 V.
2. D2 has 0.7 V on its anode and 5.0 V on its cathode: off.
3. I = (5.0 – 0.7)/4700 = 4.3/4700 = 0.91 mA (2 significant figures).

Note: A7 Diodes and LEDs/Diode Circuit Analysis
App: `N V1-0 5, 1-2 4.7k, D2-0, D2-3, V3-0 5` → `R1-2 4.7k: 4.3 V 914.9 uA 1>2 3.934 mW` *(checked)*
App: `N V1-0 5, 1-2 4.7k, D2-0, D2-3, V3-0 5` → `D2-3 .7: OFF, -4.3 V across` *(checked)*
